Section 1: Current Mirrors

2026年2月17日 星期二 14:03

Introduction

An operational amplifier has several ideal parameters, including:

- Very high differential gain
- Very low common-mode gain
- Very high input impedance
- Very low output impedance

The aim of this experiment is to explore how BJT building blocks can achieve these properties and be combined to make a working opamp.

Background

We can build 3 different stages to achieve requirements for an ideal op-amp.

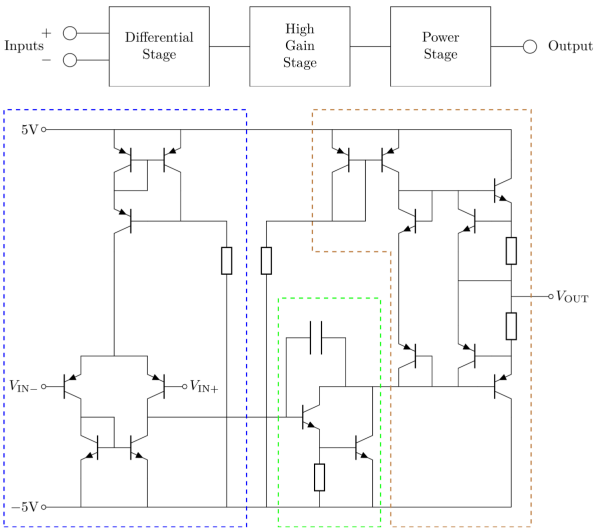


Fig1. 3 Stages in an op-amp.

The behaviour of the BJTs is defined by the usual expressions:

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}\left(1 + \frac{V_{CE}}{V_A}\right)$$

For $V_A \gg V_{CE}$, this is approximated as:

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

We will also use the small signal output impedance:

$$r_0 = \frac{V_A}{I_C}$$

Current Mirror and Current Source

The current mirror is a widely used building block that acts as a current buffer. The current flowing in the input branch of the circuit is mirrored so that the same current also flows in the output branch. An ideal current mirror has an infinite output impedance since it acts like a current source. It can be built with PNP transistors to make a source of positive current, and NPN transistors to make a current sink.

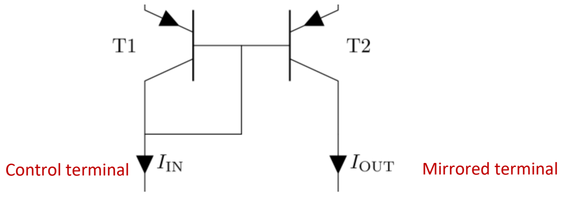
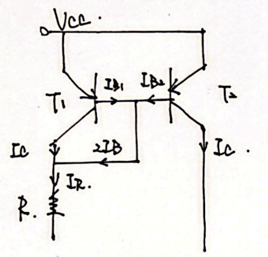


Fig2. Simple Current Mirror



$\because I_R = \frac{V_{CC} - V_{BE}}{R} = I_C + 2I_B = I_C + 2\frac{I_C}{\beta}$

$\therefore I_C = \frac{\beta}{\beta+2} I_R$.

$\because I_C$ is same for T_1 & T_2.

$\therefore$ When β is big $\Rightarrow$ $I_C \approx I_R = \frac{V_{CC} - V_{BE}}{R}$.

$I_C \approx \frac{5 - 0.7}{1 \times 10^{-3}} = 4300\Omega$

In the test 4330Ω gives 1mA.

Simulate the current mirror with VC = -15V using this circuit:

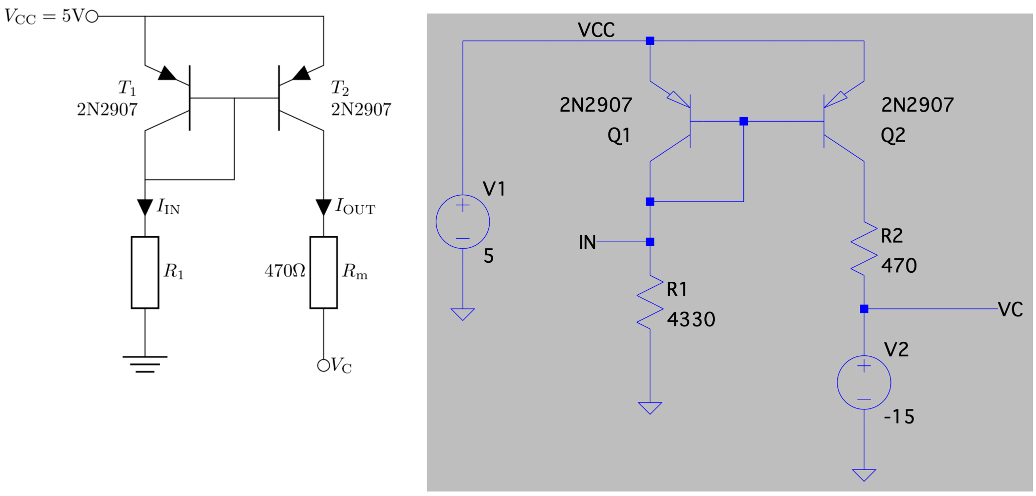
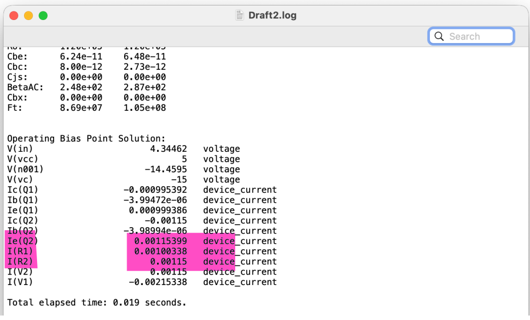


Fig3. Simulated Circuit

In this case T1 is diode-connected.

Choose a value of R_1 so that $I_{IN} = 1$ mA.



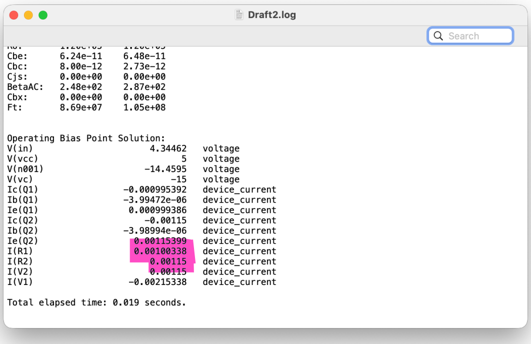
Draft2.log

```
Cbe:      6.24e-11    6.48e-11
Cbc:      8.00e-12    2.73e-12
Cjs:      0.00e+00    0.00e+00
BetaAC:   2.48e+02    2.87e+02
Cbx:      0.00e+00    0.00e+00
Ft:       8.69e+07    1.05e+08


Operating Bias Point Solution:
V(in)                     4.34462   voltage
V(vcc)                          5   voltage
V(n001)                  -14.4595   voltage
V(vc)                         -15   voltage
Ic(Q1)               -0.000995392   device_current
Ib(Q1)               -3.99472e-06   device_current
Ie(Q1)                0.000999386   device_current
Ic(Q2)                   -0.00115   device_current
Ib(Q2)               -3.98994e-06   device_current
Ie(Q2)                 0.00115399   device_current
I(R1)                  0.00100338   device_current
I(R2)                     0.00115   device_current
I(V2)                     0.00115   device_current
I(V1)                 -0.00215338   device_current

Total elapsed time: 0.019 seconds.
```

Fig4. 1mA

Run a DC analysis to find I_{IN} and I_{OUT}. What is the *current transfer ratio* between input and output?

So our IIN & IOUT is:



```
Draft2.log
Cbe:      6.24e-11    6.48e-11
Cbc:      8.00e-12    2.73e-12
Cjs:      0.00e+00    0.00e+00
BetaAC:   2.48e+02    2.87e+02
Cbx:      0.00e+00    0.00e+00
Ft:       8.69e+07    1.05e+08


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Ib(Q2)              -3.98994e-06   device_current
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I(R1)                 0.00100338   device_current
I(R2)                    0.00115   device_current
I(V2)                    0.00115   device_current
I(V1)                -0.00215338   device_current

Total elapsed time: 0.019 seconds.
```

Fig5. IOUT & IIN of Current Mirror

Ratio = IOUT/IIN = 0.00115/0.00100338 = 1.14612609

The current mirror can be compared to an ideal current source by finding its output resistance R_{OUT}. Find this by varying R_{OUT} between −15V and −5V. You can use the `DC Sweep' simulation mode, which allows you to step the value of V_{C} and record the output current for each step.

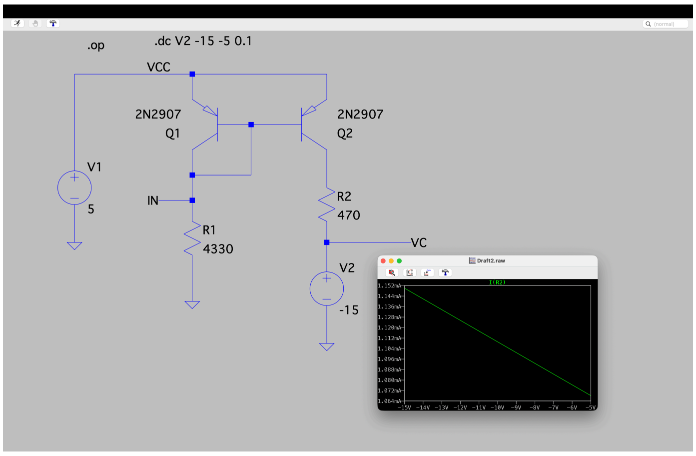


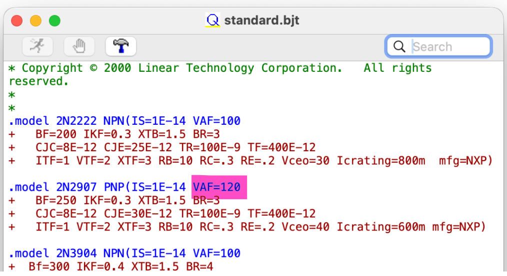
Fig6. DC Sweep

$$R_{\text{OUT}} = \frac{\Delta V_C}{\Delta I_{\text{OUT}}}$$

We get 2 points:
(-13.91V, 1.14102mA)
(-5.69V, 1.07367mA)

ROUT = 122049.0ohm

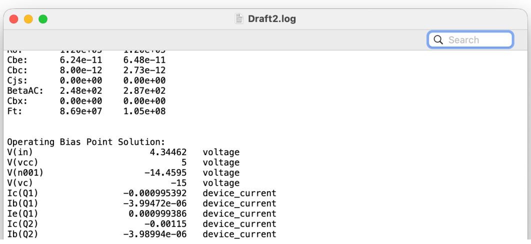
Find the output resistance of the current mirror. Does it match the predicted output resistance $r_0 = V_{\text{A}}/I_{\text{C}}$?



```
standard.bjt
* Copyright © 2000 Linear Technology Corporation.   All rights
reserved.
*
*
.model 2N2222 NPN(IS=1E-14 VAF=100
+   BF=200 IKF=0.3 XTB=1.5 BR=3
+   CJC=8E-12 CJE=25E-12 TR=100E-9 TF=400E-12
+   ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=30 Icrating=800m  mfg=NXP)

.model 2N2907 PNP(IS=1E-14 VAF=120
+   BF=250 IKF=0.3 XTB=1.5 BR=3
+   CJC=8E-12 CJE=30E-12 TR=100E-9 TF=400E-12
+   ITF=1 VTF=2 XTF=3 RB=10 RC=.3 RE=.2 Vceo=40 Icrating=600m mfg=NXP)

.model 2N3904 NPN(IS=1E-14 VAF=100
+  Bf=300 IKF=0.4 XTB=1.5 BR=4
```



```
Draft2.log
Cbe:      6.24e-11    6.48e-11
Cbc:      8.00e-12    2.73e-12
Cjs:      0.00e+00    0.00e+00
BetaAC:   2.48e+02    2.87e+02
Cbx:      0.00e+00    0.00e+00
Ft:       8.69e+07    1.05e+08


Operating Bias Point Solution:
V(in)                    4.34462   voltage
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V(n001)                 -14.4595   voltage
V(vc)                        -15   voltage
Ic(Q1)              -0.000995392   device_current
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Ie(Q1)               0.000999386   device_current
Ic(Q2)                  -0.00115   device_current
Ib(Q2)              -3.98994e-06   device_current
```
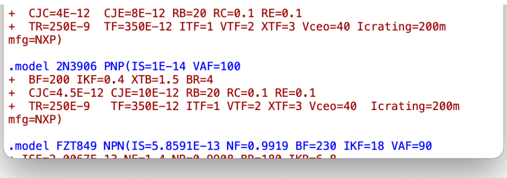
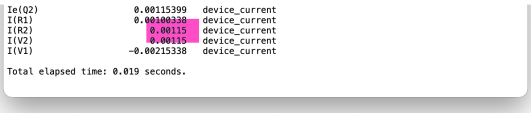



```
+  CJC=4E-12  CJE=8E-12 RB=20 RC=0.1 RE=0.1
+  TR=250E-9  TF=350E-12 ITF=1 VTF=2 XTF=3 Vceo=40 Icrating=200m
mfg=NXP)

.model 2N3906 PNP(IS=1E-14 VAF=100
+  BF=200 IKF=0.4 XTB=1.5 BR=4
+  CJC=4.5E-12 CJE=10E-12 RB=20 RC=0.1 RE=0.1
+  TR=250E-9   TF=350E-12 ITF=1 VTF=2 XTF=3 Vceo=40  Icrating=200m
mfg=NXP)

.model FZT849 NPN(IS=5.8591E-13 NF=0.9919 BF=230 IKF=18 VAF=90
```



```
Ie(Q2)                 0.00115399   device_current
I(R1)                  0.00100338   device_current
I(R2)                     0.00115   device_current
I(V2)                     0.00115   device_current
I(V1)                 -0.00215338   device_current

Total elapsed time: 0.019 seconds.
```

Fig6. Characteristic of 2N2907 in Simulating Tool

So theoretical r0 = VA/IC = 120/0.00115 = 104,347.82609ohm.
Error = 14.5%, match.

Wilson Current Mirror

The simple current mirror has a few sources of inaccuracy:

- I_C for the two transistors is only equal if the transistors are identical. In simulation this is true, but in practice transistors will have slightly different I_C for the same V_{BE}
- I_{IN} includes the I_B of T_2 and, since β is finite, this causes a difference between the currents
- BJTs have a finite output resistance and I_{OUT} will depend on V_{CE} of T_2

A much higher quality current-mirror can be obtained by the addition of just a single transistor, connected between the input and the output of the simple current-mirror. The modified circuit is shown below — it is known as the Wilson current mirror after its inventor.

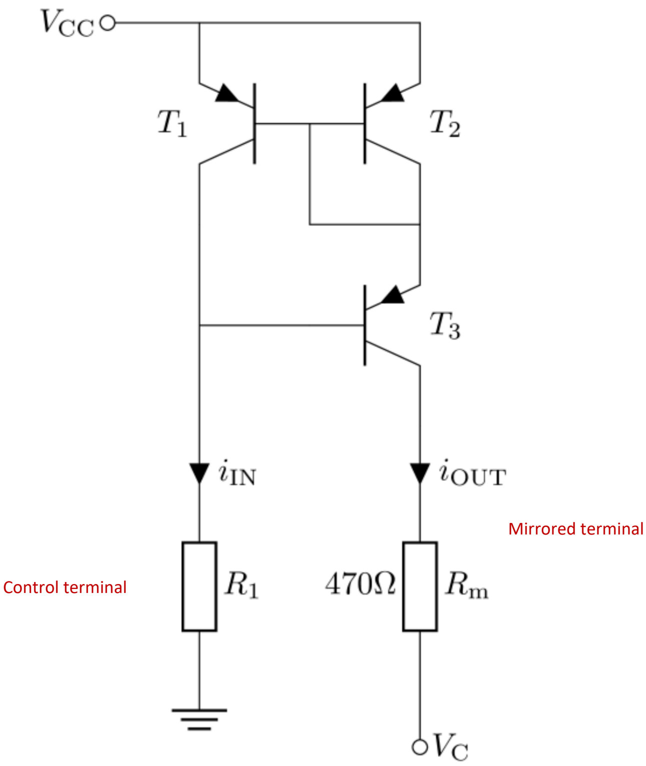
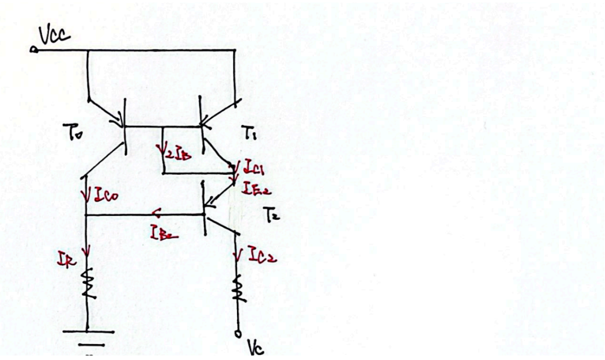


Fig7. Wilson Current Mirror

The extra transistor, T_3, now accounts for most of the voltage drop between V_{CC} and the output. That means T_1 and T_2 have a similar $V_{CE}$and therefore I_C of the two transistors is closer. T_3 also cancels out the effect of the base current in the output and reduces the error due to finite β.



$$\therefore I_{E2} = I_C + 2I_B = \frac{\beta+2}{\beta} I_C$$

$$I_C = \frac{\beta}{\beta+2} I_{E2} = \frac{\beta}{\beta+2} \cdot \frac{1+\beta}{\beta} I_{C2} = \frac{\beta+1}{\beta+2} I_{C2}.$$

$$I_R = I_{B2} + I_C = \frac{I_{C2}}{\beta} + \frac{\beta+1}{\beta+2} I_{C2} = \frac{(\beta+1)(\beta+2)}{\beta^2+2\beta} I_{C2}$$

$$I_{C2} = \left(1 - \frac{2}{\beta^2+2\beta+2}\right) I_R \approx I_R.$$

So now difference in 2 currents is a second-order difference in beta, which will be significantly reduced.

However, there is still an error because:
1. It is still dependent of beta, and beta is not infinite.
2. BJTs still have Early-Voltage resistance, so they will share the current.

Choose the value of R_1 so that $I_{IN} = 1$ mA again. Note that the voltage of the input node is now $V_{CC} - 2V_{BE}$.
Repeat the characterization of the current mirror and look for an improvement in the current transfer ratio and output impedance.

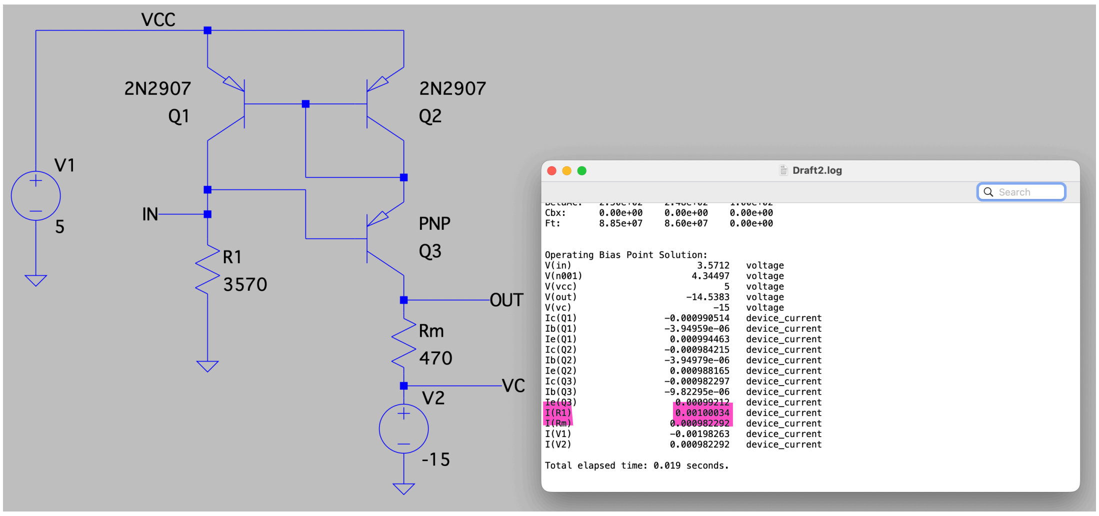


Fig8. Check For 1mA

Ratio = IOUT/IIN = 0.000982292/0.00100034 = 0.981958134.
So better than before.

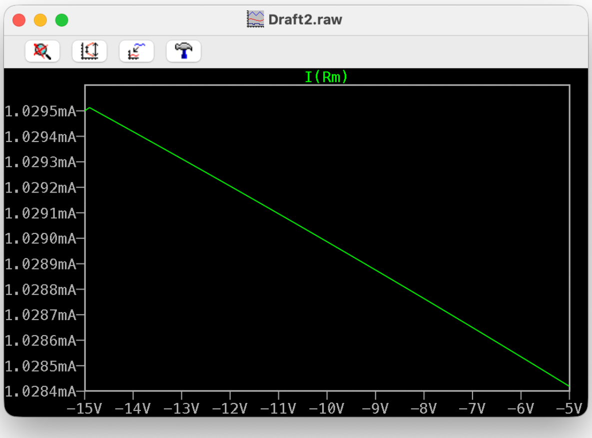


Fig9. Check for Output Resistance

(-14.26V, 1.029443mA)
(-5.68V, 1.028496mA)

ROUT = 9060190.0ohm, which is reasonable.